

3-D CINEMA BECOMES MORE REALISTIC

Experts and analysts have unanimously declared 3-D as the top emerging technology. Developments in 3-D projection technology and invention of DLP cinema technology have brought about a revolutionary change in 3-D user experience

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I remember after a long wait of nearly three weeks, I finally managed to get the tickets for Hollywood 3-D flick *Avatar*. *Avatar* has certainly marked the beginning of the 3-D era. Technology becomes more acceptable when people start to understand it. I too wanted to know how is it possible to be present in the scene with the Navis on the Pandora moon when I was not there for the shooting!

"3-D has finally come to live through *Avatar*," feels S. Ganesh, business development manager, DLP prod-

ucts, Texas Instruments. The advent of new technology to capture and project 3-D has enhanced user acceptance and experience. "New technologies have made 3-D presentation more realistic with natural look and feel," adds Ganesh.

Senthil Kumar, director and co-founder, Real Images, says "3-D is the technology of this decade. As it comes to theatre, people want it at their homes too. But for anything to happen on the small screen, it has to first happen and be successful on the big screen."

The advancement in micro-mirror technology and fast responsive software to play 3-D video has certainly enhanced 3-D user experience. You can watch 3-D without eye-fatigue or headache, which was not possible earlier. "The technology of playing the 3-D content is almost perfect today," claims Senthil.

3-D projection technologies

The 3-D projection technology was introduced way back in 1950s. But the process of setting up two cameras projecting on the same screen was extremely tedious and synchronisation was a big problem. Advancement in presentation technology with the help of digital projection has elevated stereoscopic 3-D to a quality of user experience for the first time.

There are many technologies to create and view 3-D images, primarily based on how each eye receives the

image by variable-intensity light and how the brain processes and combines the information to perceive the third-dimension depth.

"The most commonly deployed technologies in Indian theatres for 3-D are the dual projector technology, colour wheel technology and XpanD technology," informs Amitabh Vardhan, CEO, PVR Cinemas. All the technologies basically project the 3-D content containing left-eye and right-eye images on the same screen. You need to wear glasses to separate the left- and right-eye view just the way it's done naturally.

"In dual projector set-up, we have two digital projectors: one gives the left-eye view and the other the right-eye view," shares Rajiv Kumar, regional manager (engineering), PVR Cinemas. Each projector is connected to the playback server and receives the data through BNC (Bayonet Neill-Concelman) cable at the rate of 24 frames per second (fps). "Each projector has a separate filter that polarises the light and the eye-glass has the corresponding filter for each eye," explains Senthil. This is a stable arrangement, suitable for big-screen theatres and auditoria.

Another 3-D technology based on polarisation is the colour wheel. It requires a single projector. "A colour/filter wheel placed in front of the projector spins at the rate of 4320 rotations per minute (rpm)," shares Senthil. The technology patented by MasterImage 3-D uses a high-efficiency circular pro-



Colour-wheel technology (courtesy: PVR Cinemas, Delhi)